# Ryan Araula

HTTPS://rparaula.github.io/

## Profile

Analytical and detail-driven Computer Science major with a Data Science minor at the University of Houston, specializing in transforming complex datasets into actionable insights. Experienced in building data pipelines, developing interactive dashboards, and applying machine learning models to real-world problems. Known for strong problem-solving skills, clear communication, and the ability to quickly learn new tools and technologies. Seeking an opportunity to contribute to a data-driven team and grow within the fields of Data Engineering, Analytics, or Machine Learning.

## Education

### University of Houston

Bachelor's in computer science, Minor in Data Science
August 2021 — May 2026

Certifications

- Microsoft Azure AI Fundamentals
- Microsoft Azure Data Fundamentals
- Tableau Desktop Specialist
- Microsoft Power BI Data Analyst Associate (Pending)
- AWS Certified Cloud Practitioner (Pending)

## Projects/Experience

### Postal Office Database Application

- Designed and implemented a MySQL relational database modeling customer, worker, and manager workflows for a postal system.
- Developed a full-stack web application with secure login, user-specific dashboards, and data entry/reporting tools using JavaScript..
- Deployed the application and database on Microsoft Azure, demonstrating cloud-based hosting, scalability, and team collaboration.

### COVID Outbreak Data Analysis

- Conducted exploratory data analysis on global and U.S. COVID-19 datasets to uncover trends, anomalies, and missing data patterns.
- Engineered a unified dataset using feature engineering (rolling averages, ICU/hospitalization rates, case-fatality metrics).
- Built regression-based models to classify high-risk periods for individual countries, showcasing statistical modeling and data storytelling.

### Phishing Email Classifier + Adversarial Machine Learning

- Consolidated multiple email corpora into a clean, standardized multi-class dataset (ham, spam, phishing) for pre- and post-2010 samples.
- Built two machine learning pipelines combining TF-IDF NLP features with RDAP/DNS metadata to improve adversarial robustness.
- Evaluated model performance across datasets and stress-tested pipelines using TextAttack, analyzing model complexity vs. vulnerability.

### Automated Air Quality Data Collection for ML Pipeline

- Built a Python pipeline automating daily collection of air quality, weather, and pollution source data across 6 APIs into a structured dataset for ML-based AQ prediction in Houston
- Implemented incremental state tracking and metadata deduplication to ensure idempotent daily runs with automatic crash recovery

## Details

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DATE / PLACE OF BIRTH
05/17/2003
Pearland, TX

## Languages

C++; C; Python; Latex; Javascript; SQL

## Libraries

Pandas; Matplotlib; NumPY; Sckit-Learn; Seaborn;

## Tools & Platforms

Microsoft Office; Visual Studio; Microsoft SQL Server; Tableau; Azure; Power BI; Jupyter Notebooks; Google Colab

## Modeling Techniques

Exploratory Analysis; Statistical Analysis; Data Mining; Linear Regression; Logistic Regression; Support Vector Machines